GATE-AII BRANCHES
Engineering Mathematics



Single variable calculus

Lecture No.- 01



By- Rahul sir

Topics to be Covered



Topic	Basic graphs

Calculus :-

Single Variable Calculus :

15-20 marks

3-4 marks

→ EE/EC/CE/ME/PI

CM

CS - 2 marks

Pakka

All branches

(10) Max/min of a Function

1) Functions — graphs Basics

2) Differentiation / Jacobian / mean value Theorem

3) Limit and continuity

4) Differentiability

5) Integration + Definite Integration

6) Taylor SERies / maclurins Series

(7) Area Bounded regions

(8) Length of curve

(9) Special integrals / Improper Integral

Beta Gamma

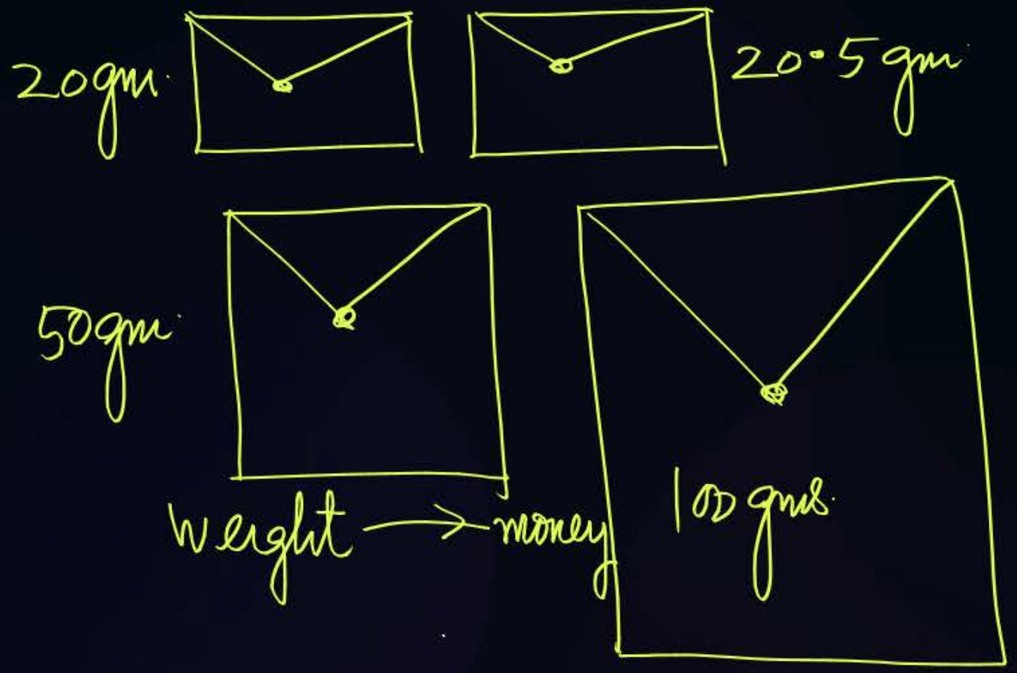
Basic graphs:

Function: $y = f(x)$



x = Independent variable

y = Dependent variable



Weight → Money ₹

- 20 gm → 1 ₹
- 20·5 gm → 1.05 ₹
- 50 gm → 5 ₹
- 100 gm → 10 ₹

Weight → money

Dependence

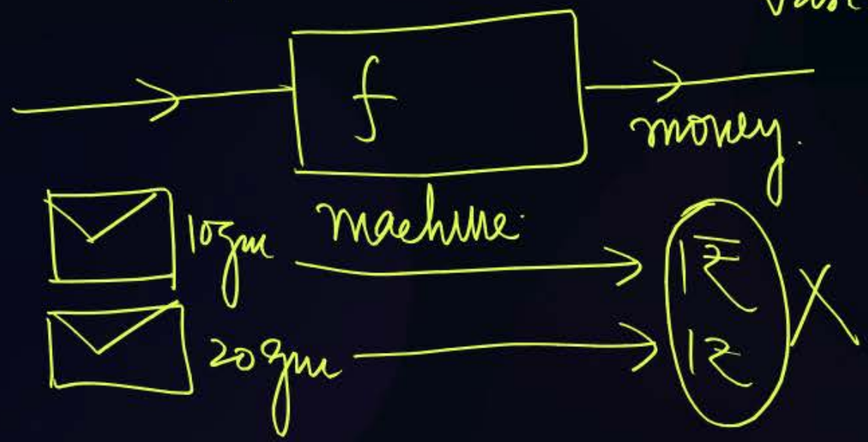
Relationship

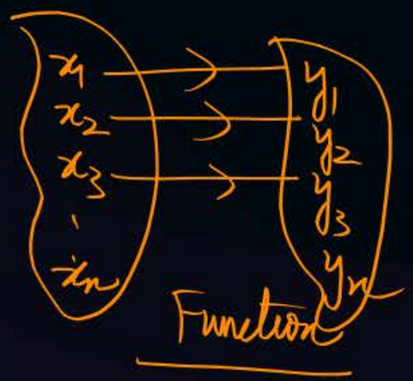
Money = f(weight)



Y = money
variable
Dependent

X = weight
Independent
variable



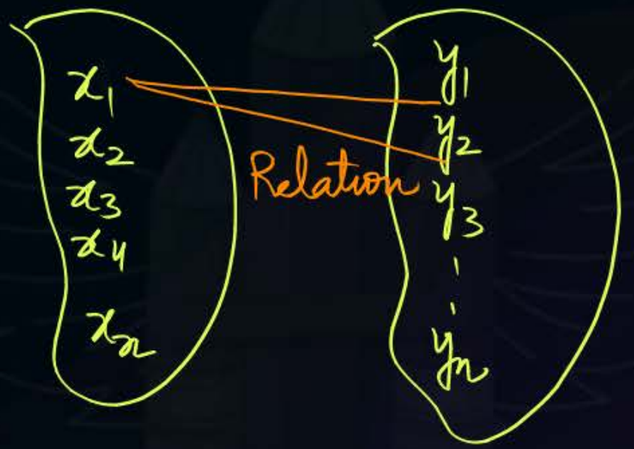


f → process

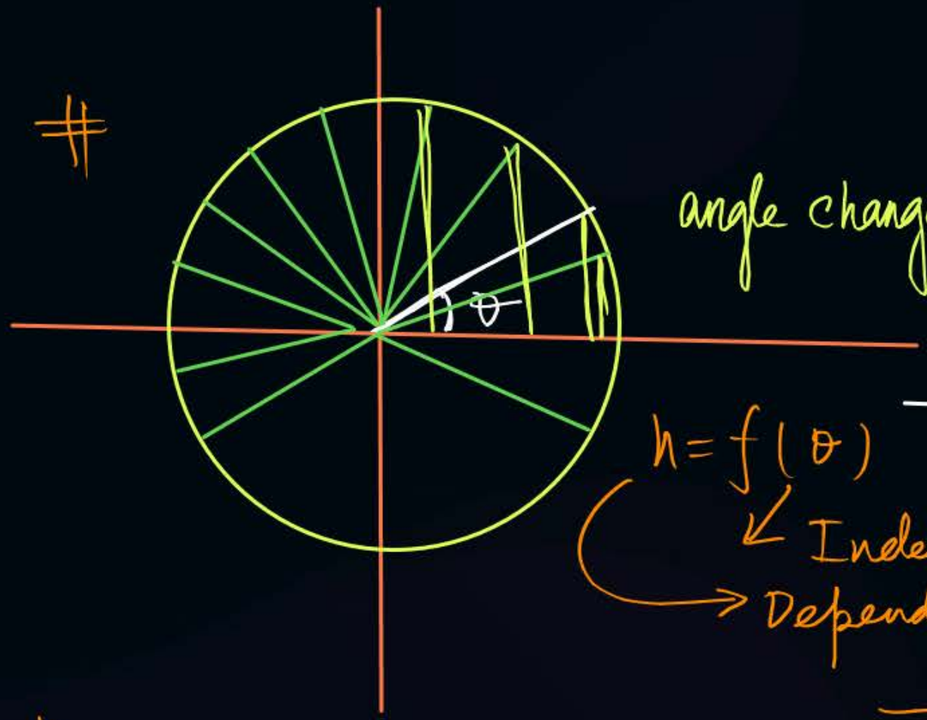
Certain Rule

" A Rule between Dependent or Independent variable"

$y = f(x)$



#



angle change ⟶ Height change
⟶ Vertical Distance

$h = f(\theta)$

Angle θ change.

$h = f(\theta)$ ⟶ Height also change.

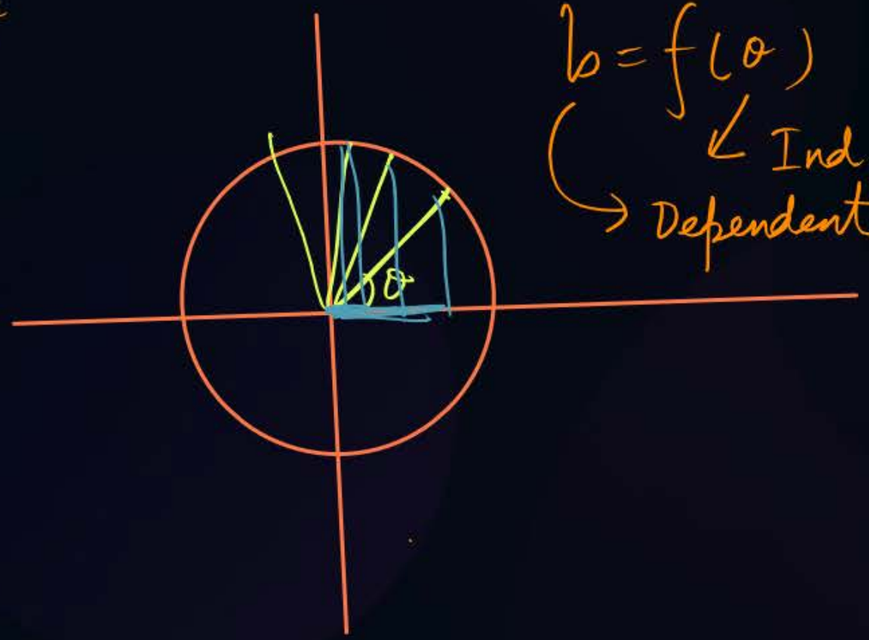
Vertical Distance

$h = f(\theta)$

↙ Independent

↳ Dependent

#



$b = f(\theta)$

↙ Ind.

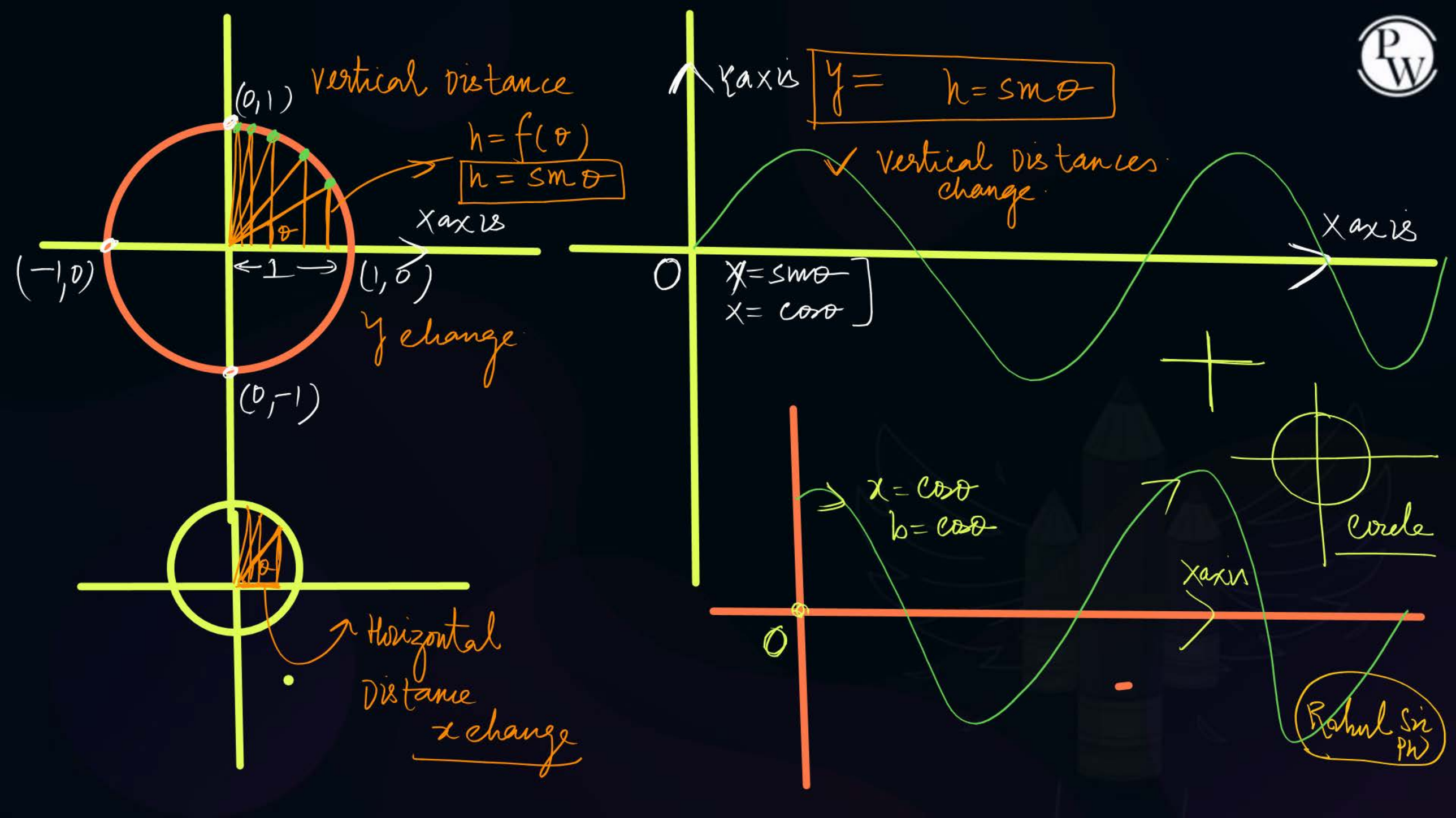
↳ Dependent

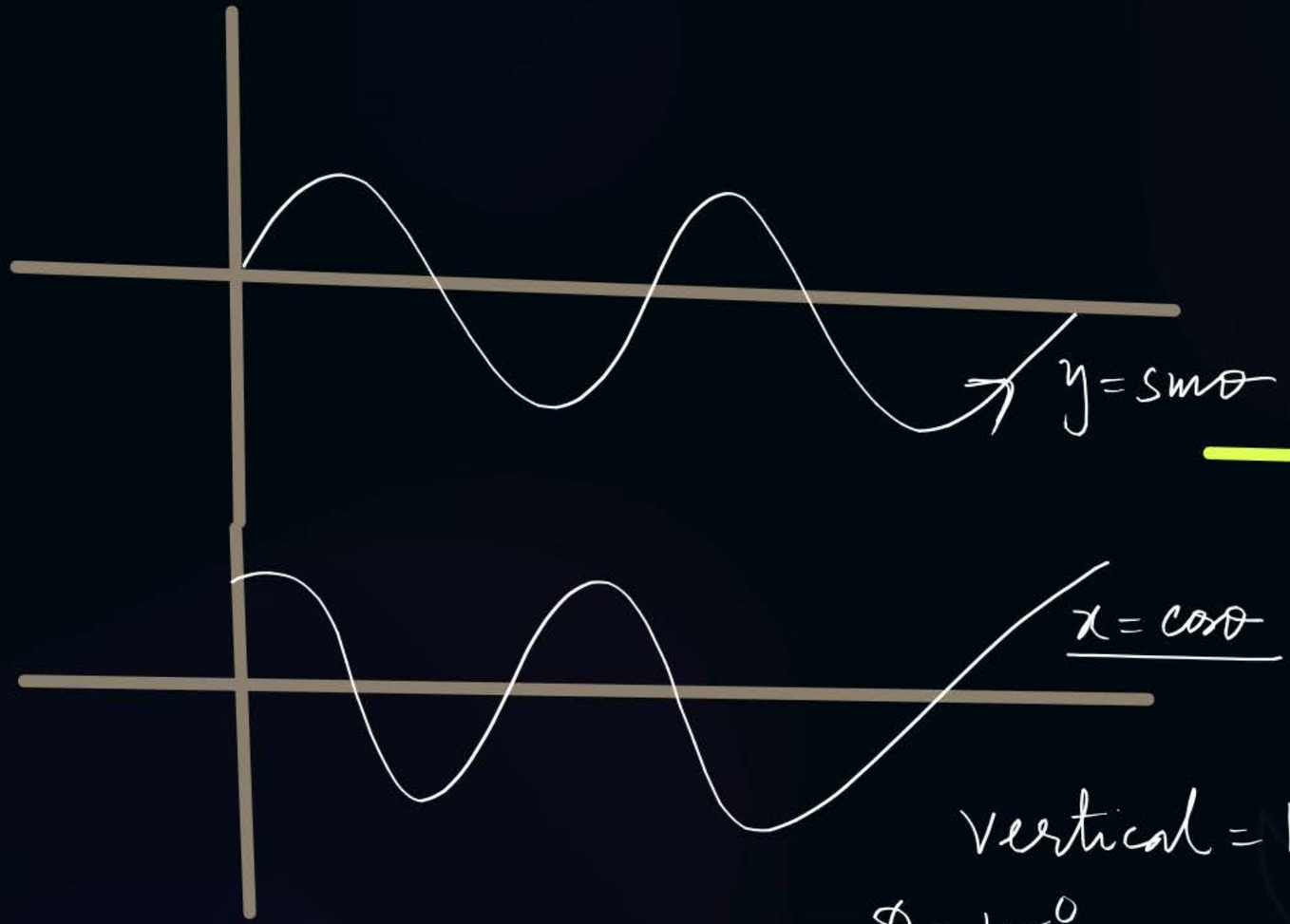
Angle θ change.

$b = f(\theta)$ ⟶ Height also change.

Horizontal Distance

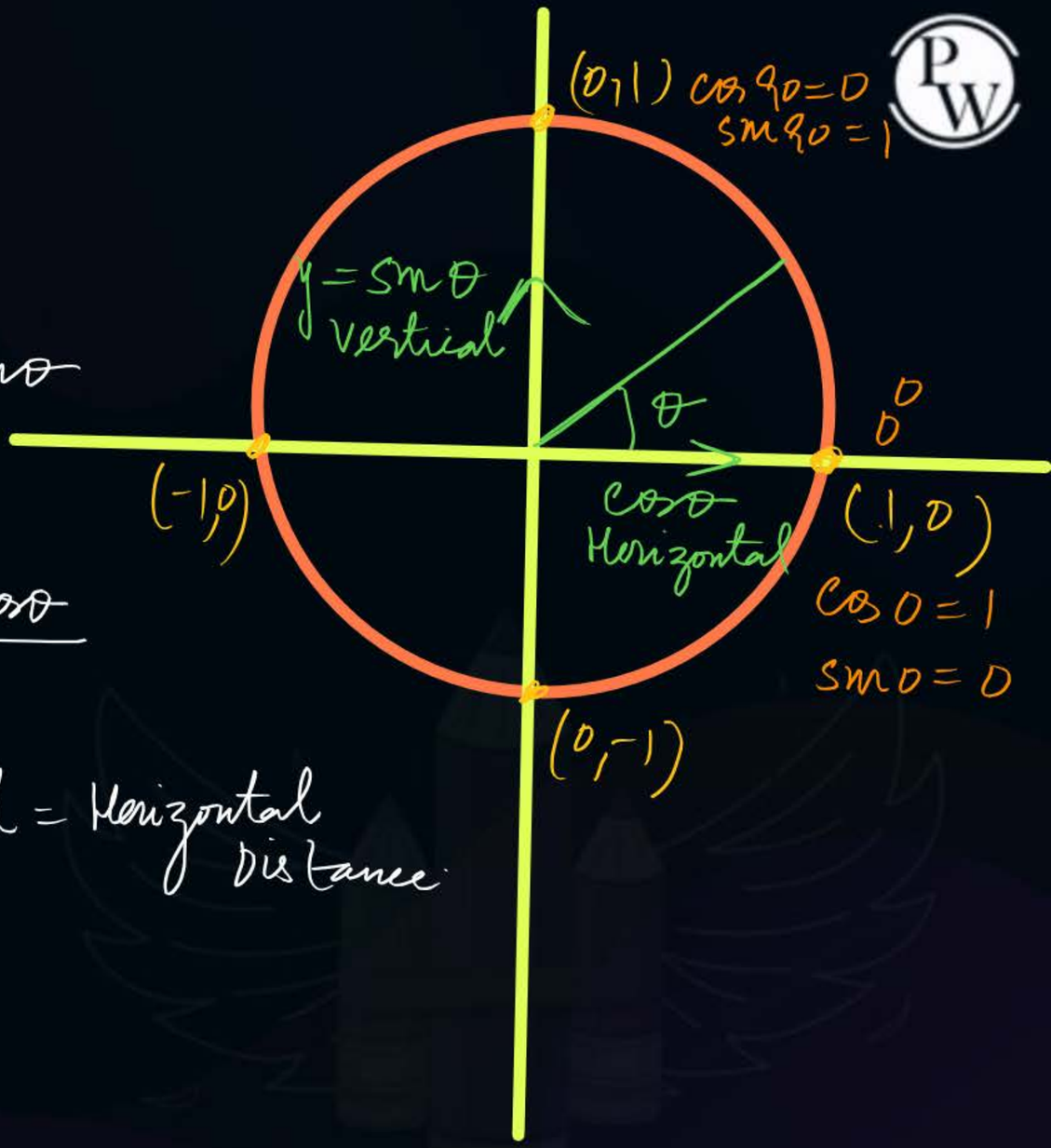
Rule of functions


(0,1)
Vertical Distance
h = f(θ)
h = sin θ
θ
x axis
(−1,0)
1
(1,0)
y change
(0,−1)
b
Horizontal
Distance
x change
y axis
y = h = sin θ
Vertical Distances change
x axis
O
x = sin θ
x = cos θ
+
x = cos θ
b = cos θ
O
x axis
-
Circle
Rahul Sir PW



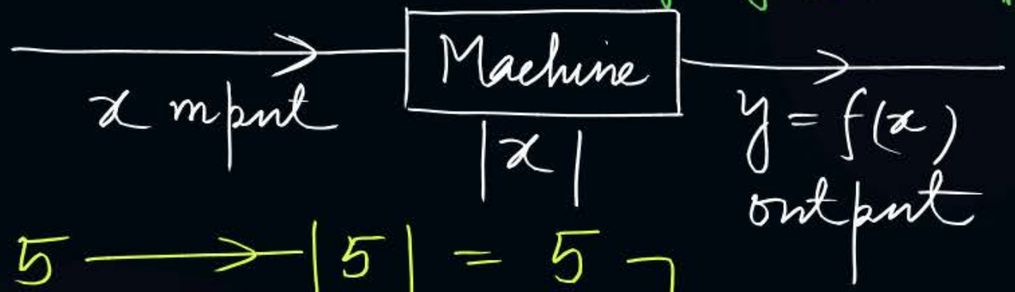
Vertical = Horizontal Distance

$\theta = 45^\circ$



Modulas function:

$y = f(x) = |x|$ = always positive



always Positive

$5 \longrightarrow |5| = 5$
$2 \longrightarrow |2| = 2$
$3 \longrightarrow |3| = 3$

$-1 \longrightarrow |-1| = 1$
$-2 \longrightarrow |-2| = 2$
$-3 \longrightarrow |-3| = 3$

output always positive

$y = |x|$

- x $x \geq 0$
- $-x$ $x < 0$

y = compound Function

- mix ① x $x \geq 0$
- mix ② $-x$ $x < 0$

$$f(x) = y = |x| = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$$

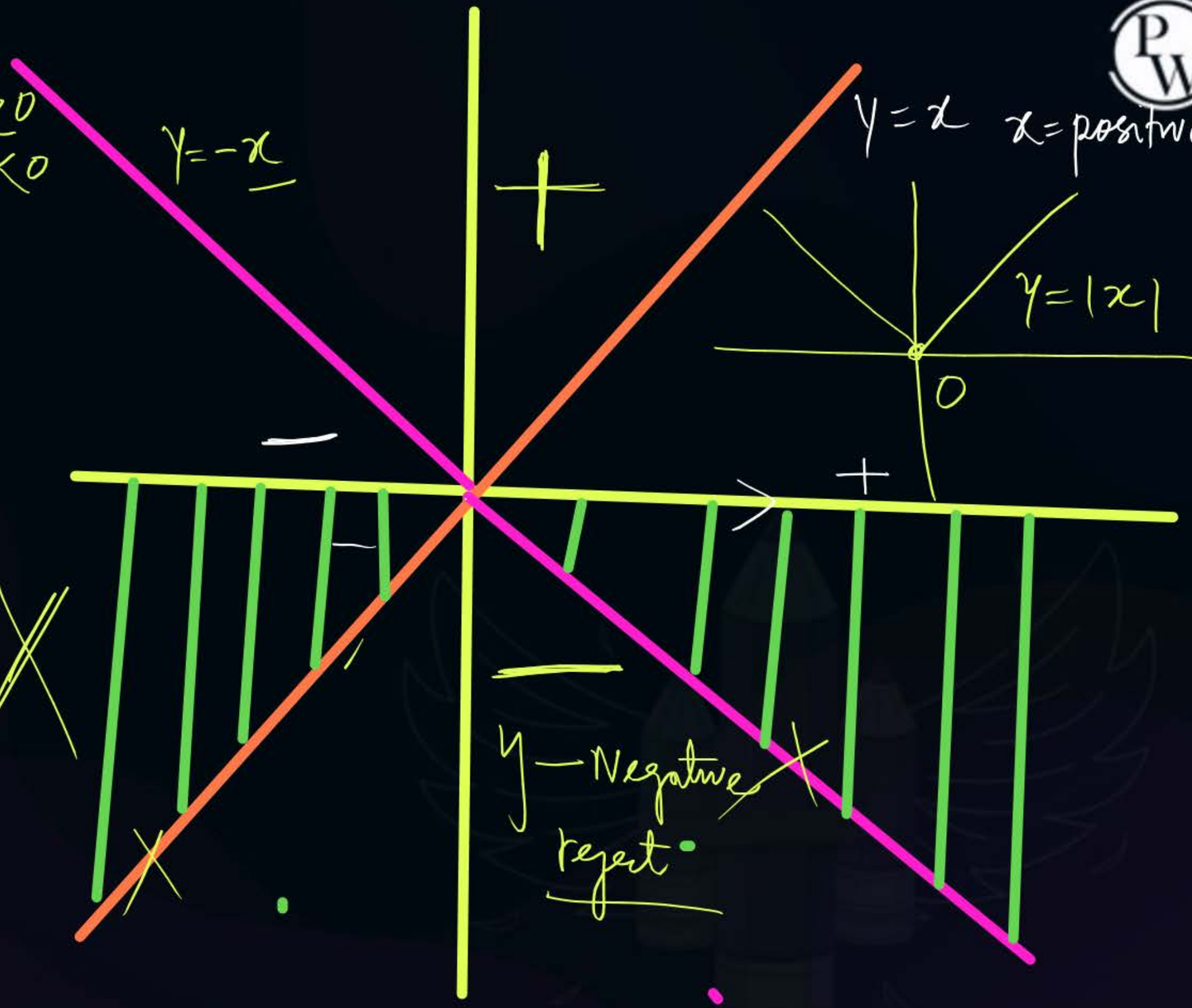
Equation of Line

$y = mx$, $y = x$

$m = 1$

$\boxed{\theta = 45^\circ}$

$y = -x \quad \theta = 135^\circ$



Transformation

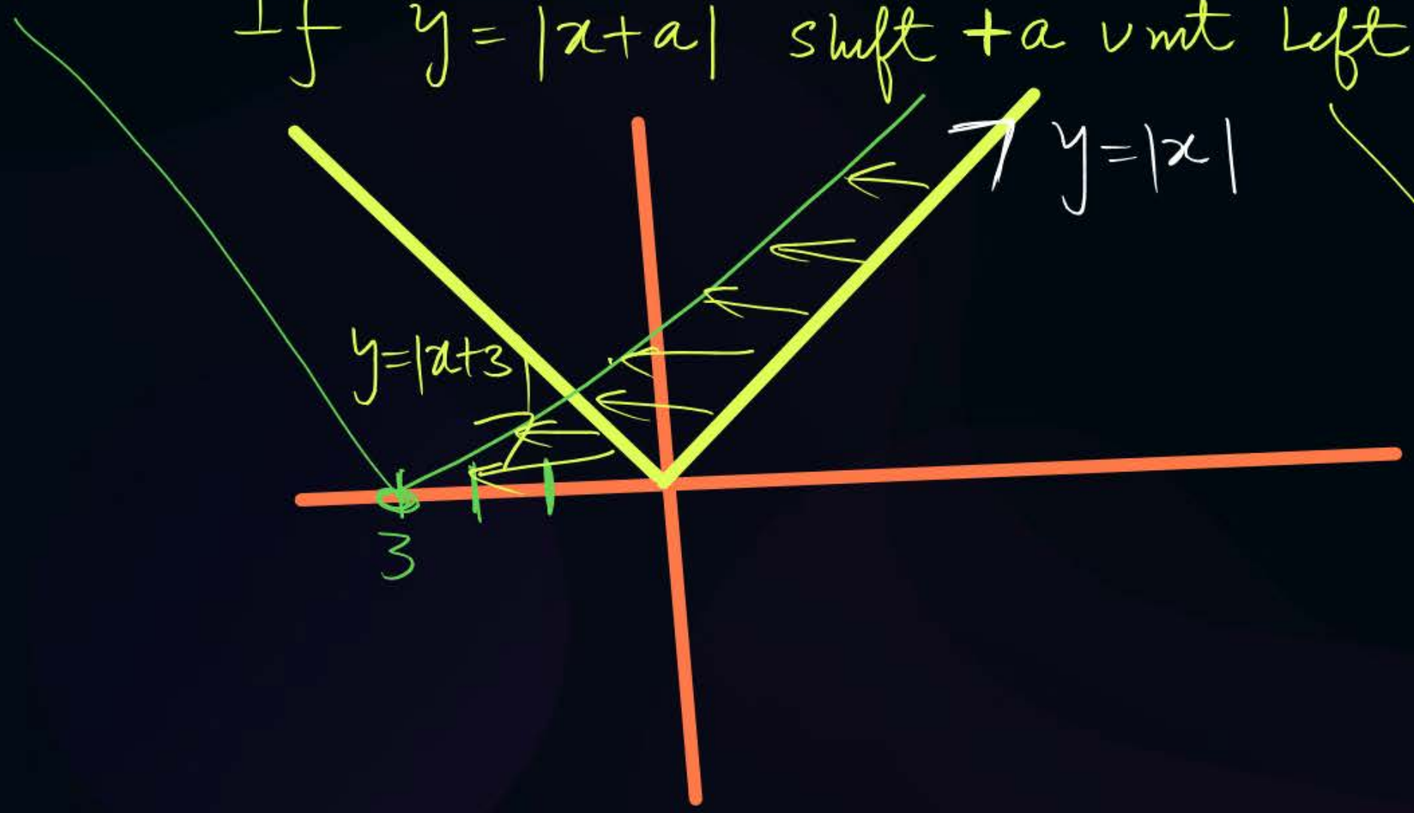
$y = |x+3|$ sketch The curve

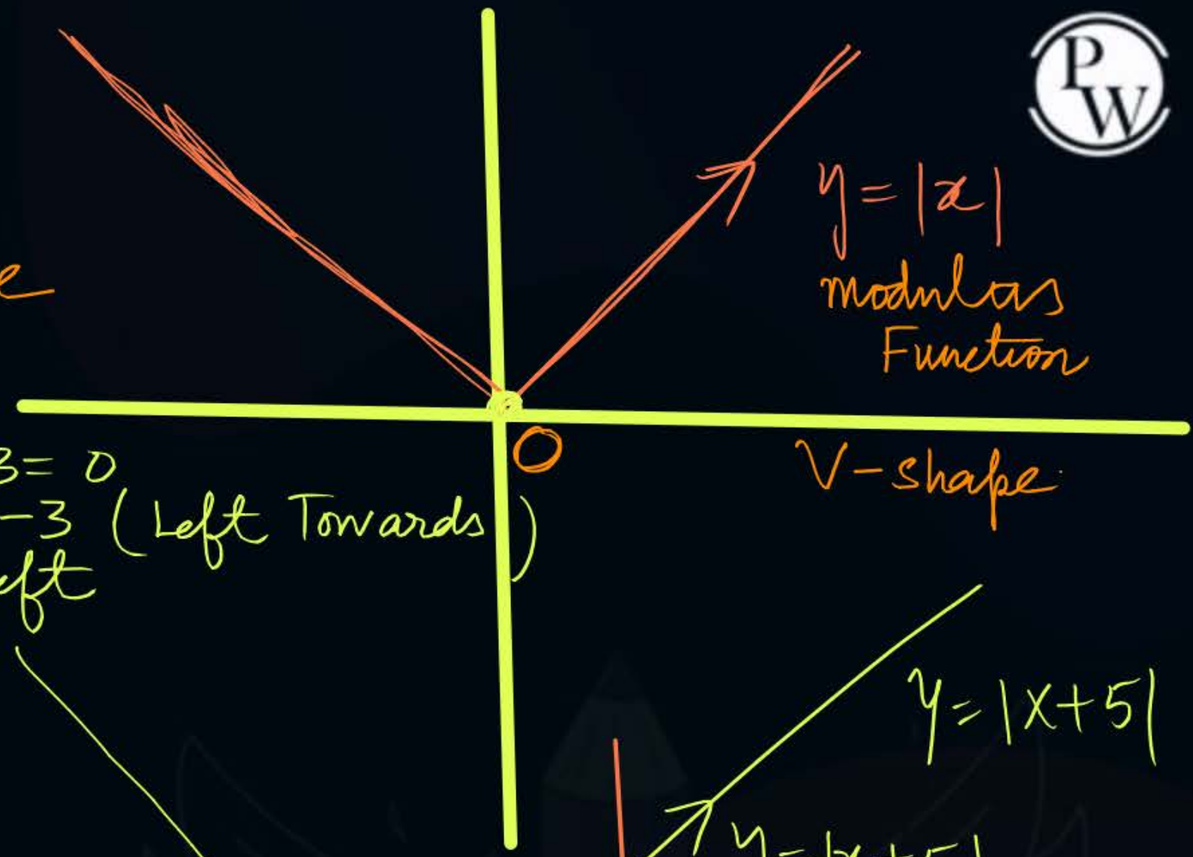
$y = |x-3|$

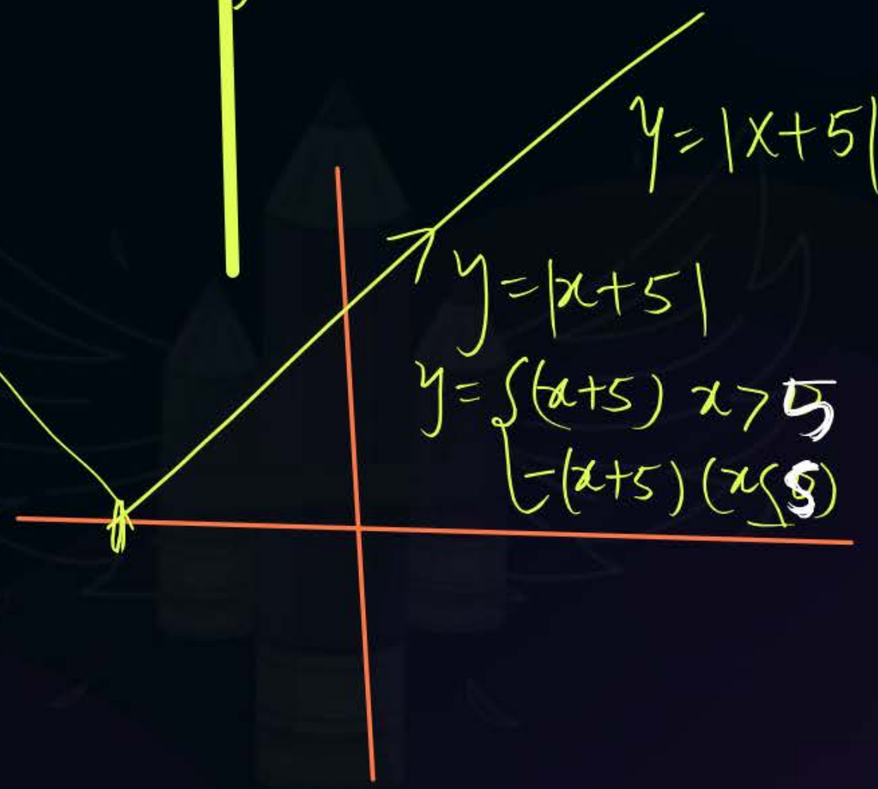
Rule $y = |x+3| = |f(x)|$ $f(x) = x+3 = 0$

$x = -3$ (Left Towards)

If $y = |x+a|$ shift $+a$ unit Left







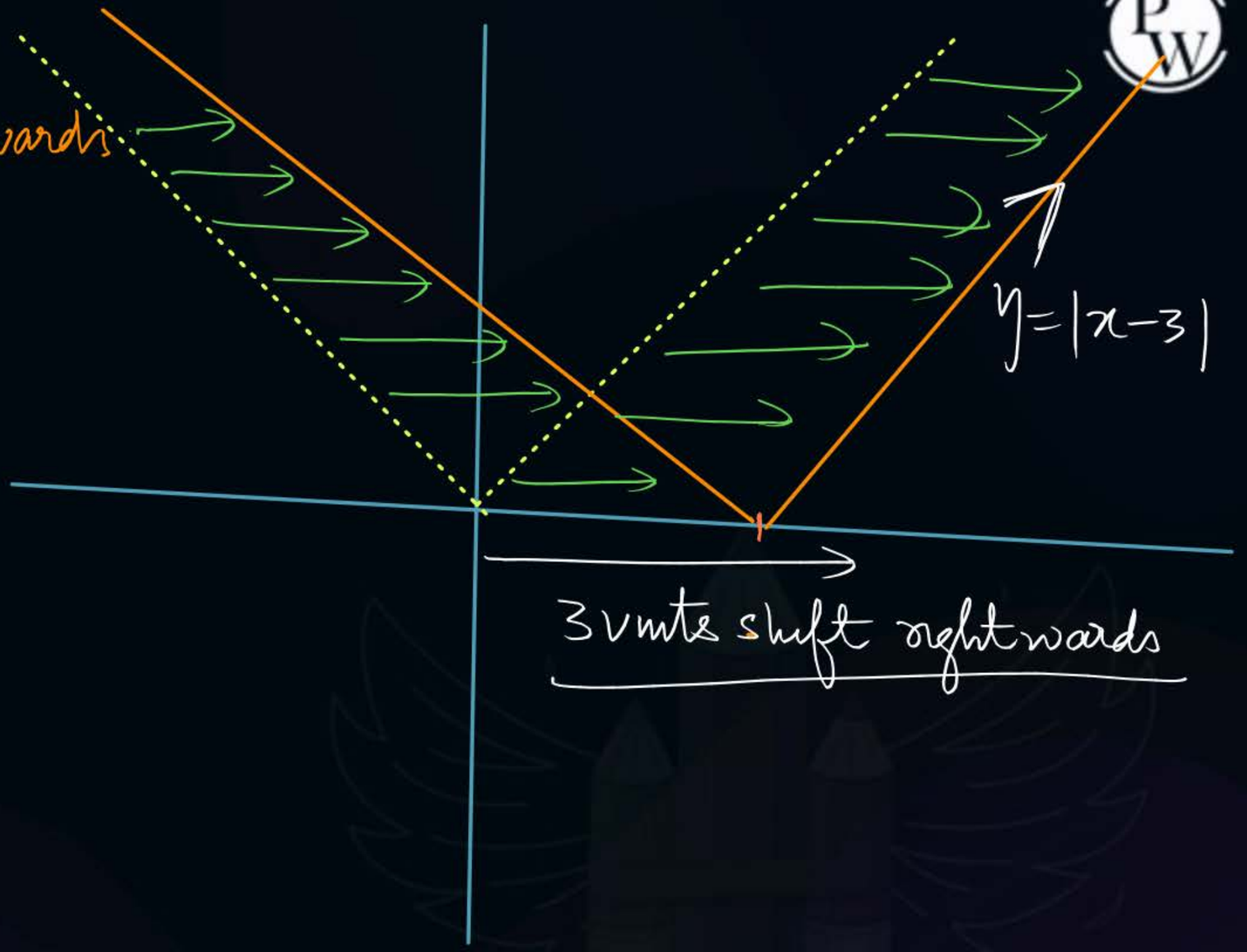
$y = |x-3|$

$y = |x-a|$ "a unit shift rightwards Direction"

Sketch the graphs

$y = |x+5|, |x-5|$

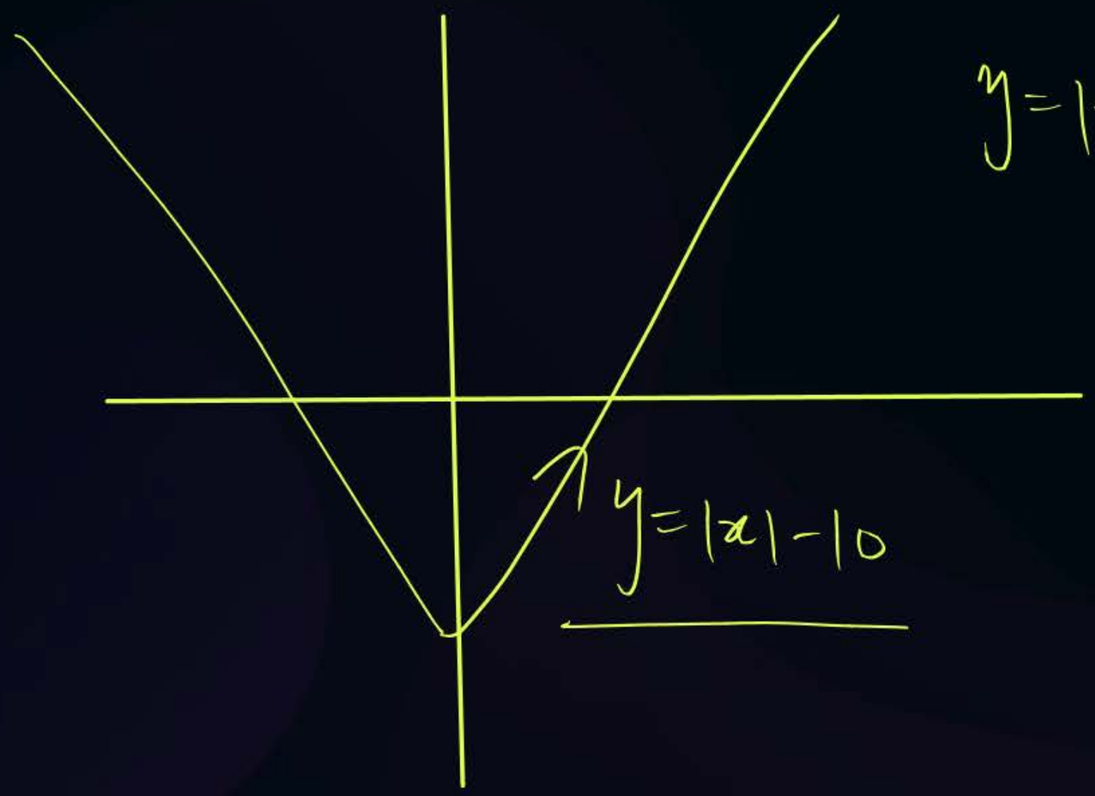
$y = |x+10|, |x-10|$



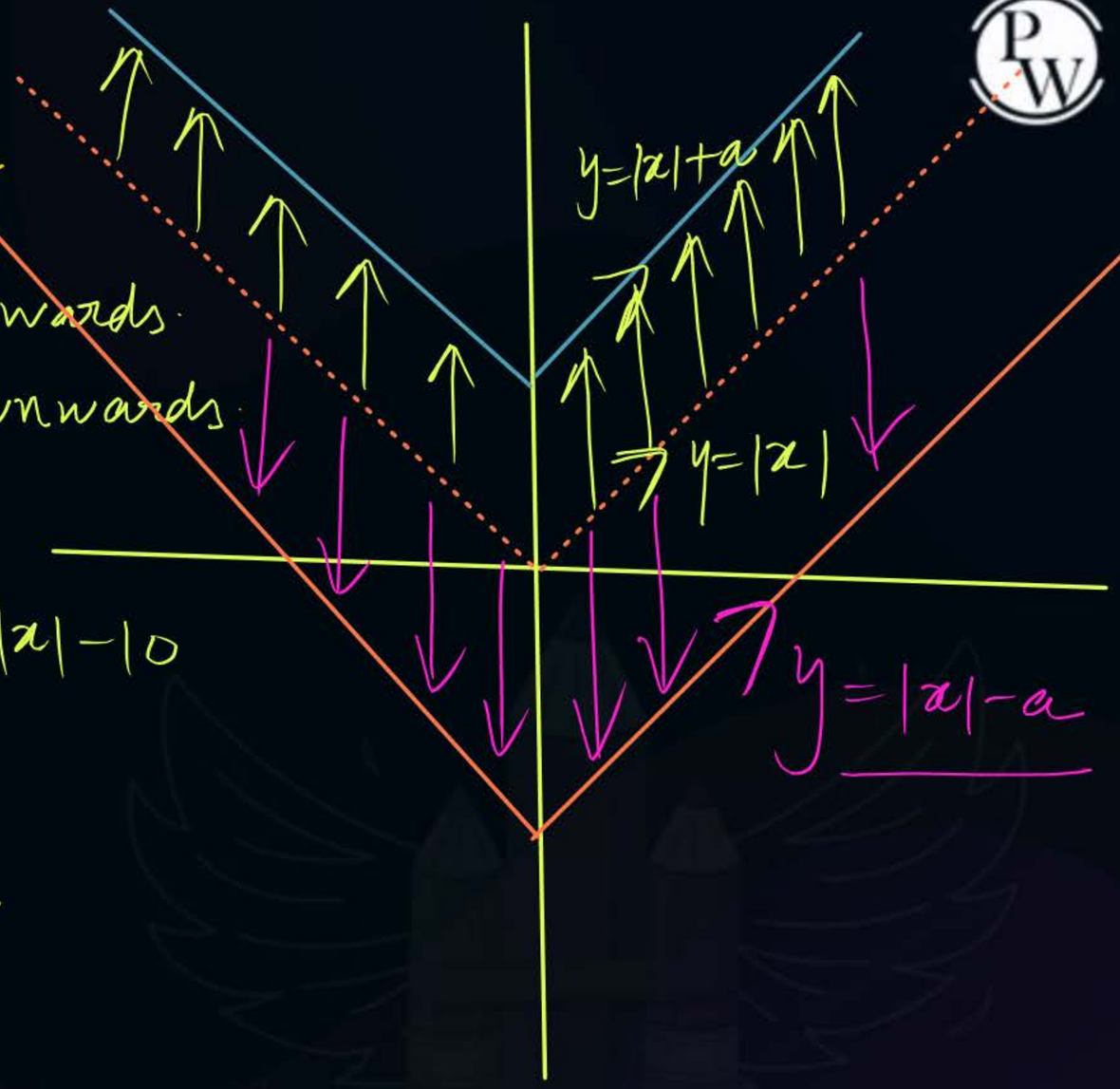
CASE 02 $y = f(x) = |x| \pm a$

" $y = |x| + a$ a unit ↑ upwards.

" $y = |x| - a$ a unit ↓ downwards.

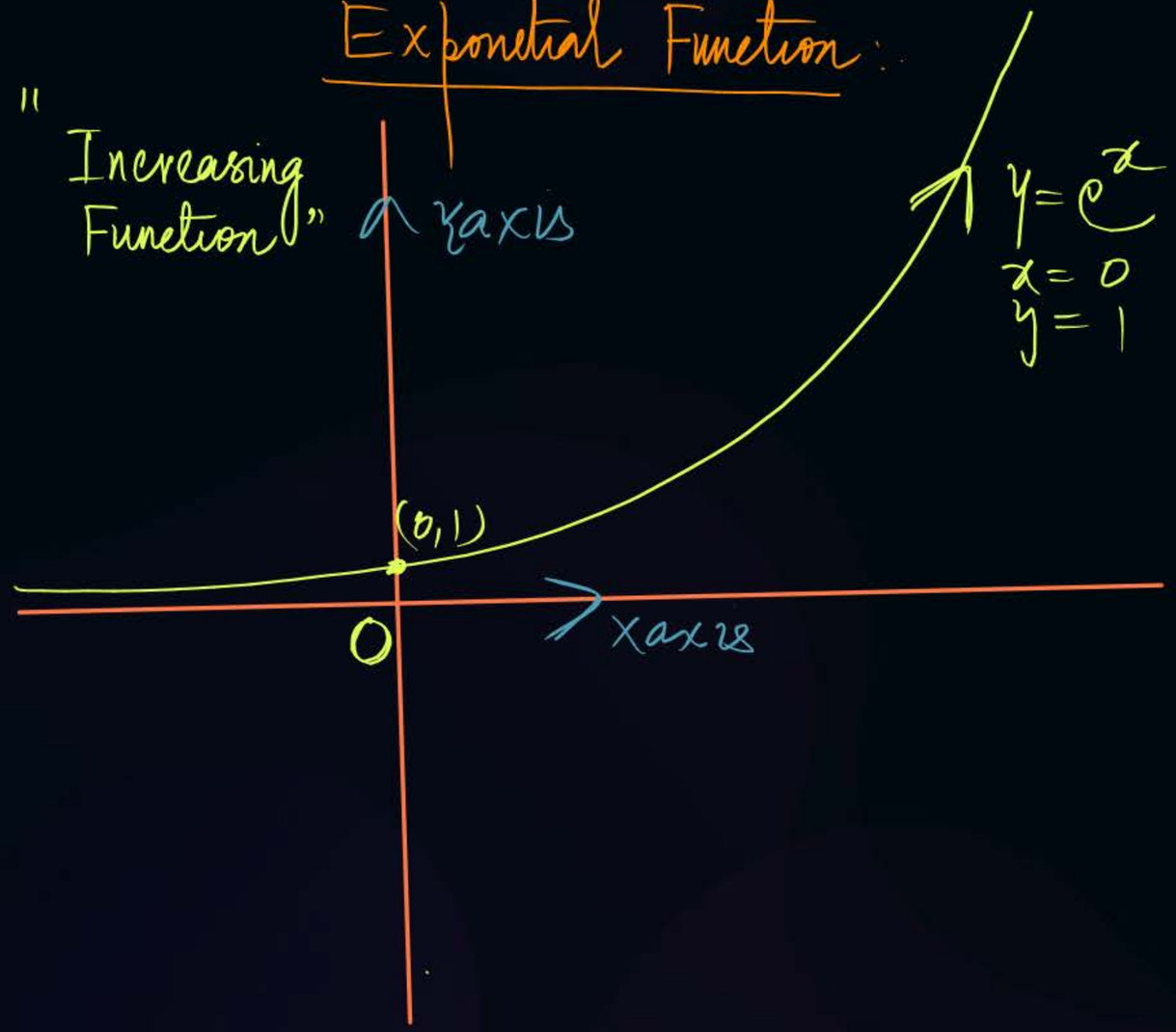


$y = |x| - 10$

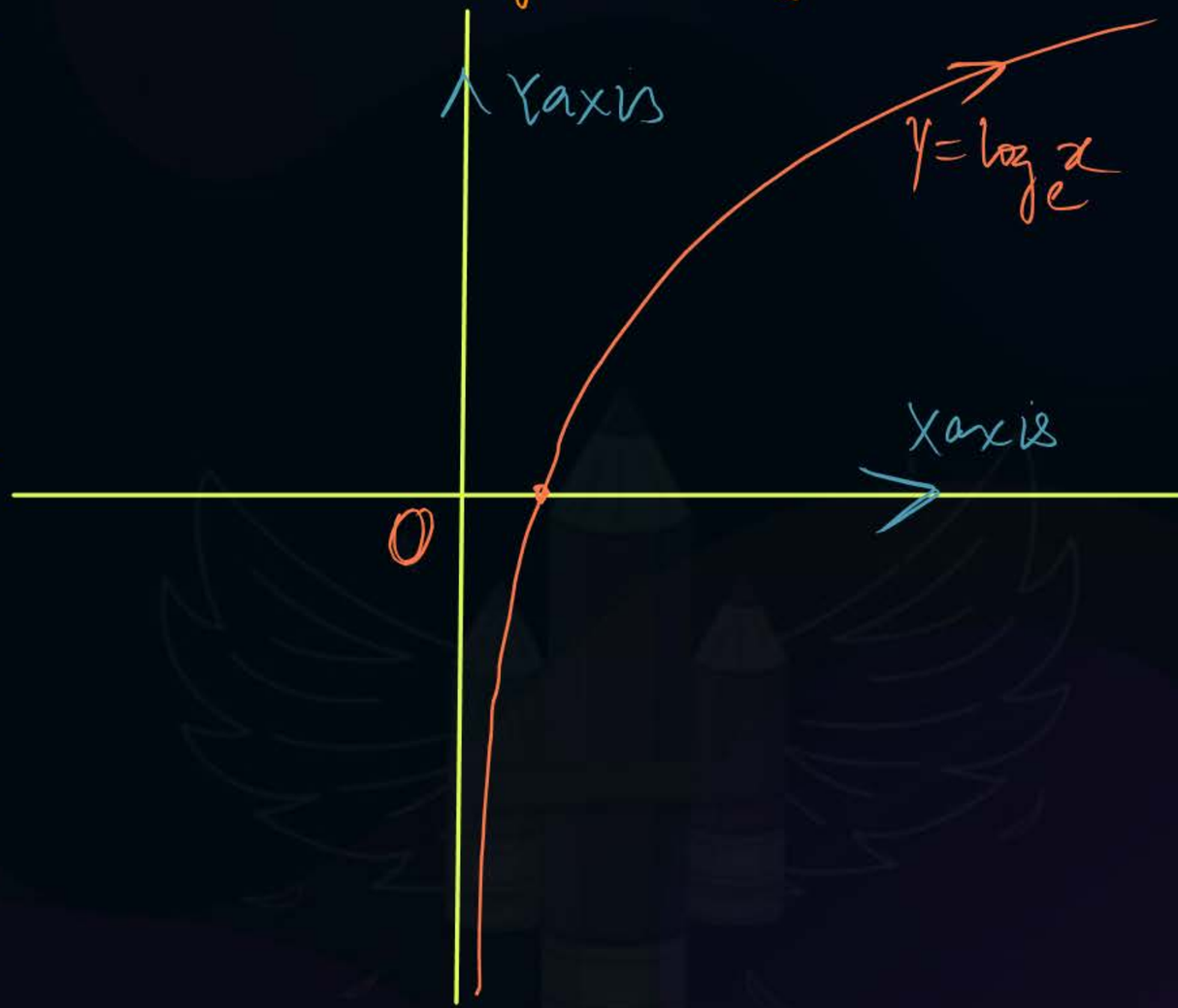


Exponential Function:

"Increasing Function"



Logarithmic function



16 Nov (+) (2 hrs)

THANK - YOU